

# 10

## FILE FINDER



In this chapter, you'll write your own recursive program to search for files according to custom needs. Your computer already has some file-searching commands and apps, but often they're limited to retrieving files based on a partial filename. What if you need to make esoteric, highly specific searches? For example, what if you need to find all files that have an even number of bytes, or files with names that contain every vowel?

You likely will never need to do these searches specifically, but you'll probably have odd search criteria someday. You'll be out of luck if you can't code this search yourself.

As you've learned, recursion is especially suited to problems that have a tree-like structure. The filesystem on your computer is like a tree, as you saw back in Figure 2-6. Each folder branches into subfolders, which in turn can branch into other subfolders. We'll write a recursive function to navigate this tree.